SYNOPSIS OF MINI PROJECT ON

"BLOCK CHAIN BASED SUPPLY CHAIN TRANSPARENCY FOR AGRICULTURAL PRODUCE"

"AGRICRED"

Submitted In Partial
Fulfillment Of The Requirements For The Award Of The Degree Of

BACHELORS OF TECHNOLOGY
IN
COMPUTER SCIENCE & ENGINEERING

By

Kavya Johari (2400680100260)
Keshav Agarwal (2400680100263)

Under Guidance of

Faculty Mentor Name
Navneet Gupta

Department of Computer Science and Engineering
Meerut Institute of Engineering and Technology,
Meerut
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Declaration

We hereby declare that the synopsis of project **entitled " BLOCK CHAIN BASED SUPPLY CHAIN TRANSPARENCY FOR AGRICULTURAL PRODUCE "** to be submitted for the Degree of Bachelor of Technology in Computer Science and Engineering is our original work and the synopsis has not been submitted for the award of any degree, diploma, or fellowship of similar other titles in previous work. It has not been submitted to any other University or Institution for the award of any degree or diploma.

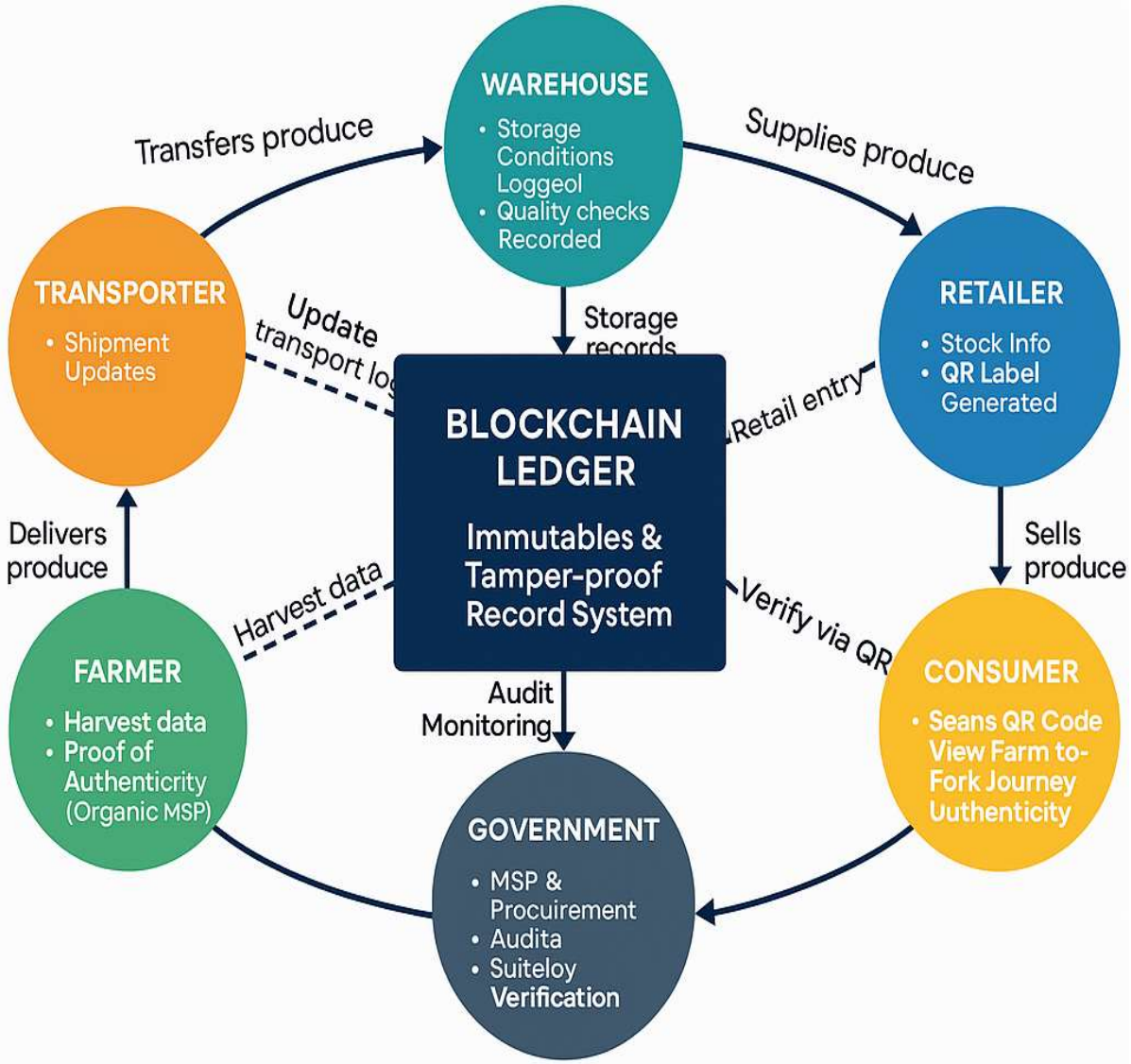
Signature and detail of student
Kavya Johari (2400680100260)
Keshav Agarwal (2400680100263)

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Blockchain-Enabled Farm-to-Fork Supply Chain Transparency



1. Introduction

Agriculture forms the backbone of India's economy, yet a lack of transparency in the agricultural supply chain causes severe issues such as price manipulation, adulteration, and unfair trade practices. Middlemen exploit farmers, consumers are unable to verify the authenticity or quality of produce, and government subsidies such as MSP are often misused due to tampered records.

To address this, a transparent, trustworthy, and tamper-proof system is required.

Blockchain technology provides immutable records, decentralization, and traceability, making it an ideal solution for monitoring agricultural transactions throughout the supply chain—from farms to consumers. This project proposes a blockchain-based platform that enhances transparency, authenticity, and efficiency in agricultural supply management.

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2. Problem Statement

The existing agricultural supply chain does not provide adequate transparency regarding pricing, quality, or provenance. Key issues include:

- Exploitation of farmers due to middlemen-driven pricing.
- Consumers cannot validate the origin or quality of produce.
- Manipulation and misuse of MSP/subsidy records.
- Lack of traceability of quality certificates and logistics data.

There is a pressing need for a system where produce details are transparently recorded, verified, and accessible to all stakeholders.

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3. Objectives

The proposed system aims to:

- Ensure tamper-proof recording of agriculture supply chain events using blockchain.

- Provide proof of authenticity, origin, quality certification, and fair pricing.
- Enable government agencies to track MSP/subsidy allocation transparently.
- Empower consumers with QR-based product verification.
- Improve trust and accountability among farmers, logistics partners, retailers, and consumers.

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4. Proposed Solution

The system will record every event in the agricultural supply chain (harvest, transportation, warehouse storage, retailing) using **blockchain smart contracts** deployed on **Polygon PoS (Layer-2)**. Quality certificates, invoices, and logistic receipts will be stored on **decentralized IPFS storage**, and their hash values referenced on the blockchain to prevent tampering.
A **web-based platform (Next.js)** will provide dashboards for farmers, distributors, government officials, and retailers. A **QR Code** will be generated for each produce batch, enabling consumers to verify authenticity and journey details.

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5. Key Features

Feature	Description
Blockchain Smart Contracts	Record supply chain transitions immutably
IPFS Integration	Decentralized storage for certificates and receipts
QR Code Traceability	Scan-and-verify product journey & origin
Government Dashboard	Real-time MSP/subsidy tracking & compliance
Hybrid Architecture	Blockchain + MongoDB for low-cost scalability
Web-First Design	No mobile app required, accessible to all

6. System Architecture

Frontend: Next.js + TailwindCSS + Shadcn/UI
Backend: Node.js + Express.js
Blockchain: Solidity Smart Contracts on Polygon PoS
Database: MongoDB (off-chain structured data)
Storage: IPFS (decentralized document storage)
Connectivity: Ethers.js + WAGMI Wallet Integration

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7. Methodology / Workflow

1. **Farmer uploads harvest details & quality documentation.**
2. **Warehouse verifier checks authenticity & updates storage logs.**
3. **Transport partner logs shipment temperature/humidity data.**
4. **Retailer verifies stock and updates sale records on blockchain.**
5. **QR code generated for customers to scan and validate authenticity.**
6. **Government dashboard analyzes MSP/subsidy disbursement, compliance, and fraud detection.**

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8. Feasibility Study

- **Technical Feasibility**
  - Polygon PoS scalability, low gas cost; hybrid database enables performance.
- **Operational Feasibility**
  - QR-based end-user flow, multilingual support, suitable for rural deployment.
- **Financial Feasibility**
  - Low-cost open-source tools, Layer-2 blockchain reduces cost, increases farmer income and reduces fraud.

9. Expected Outcomes

- Transparent and traceable agricultural supply chain.
- Increased farmer income and reduced exploitation.
- Consumer trust through verified organic/local claim tracking.
- Reduced subsidy and pricing fraud for government agencies.
- Efficient tracking of quality, logistics, and produce journey.

10. Impact

Stakeholder	Benefit
Farmers	Fair pricing, digital trust, authentic recognition
Consumers	Verified quality & authenticity through QR
Government	Transparent subsidy/MSP distribution
Supply Chain Actors	Reduced spoilage, accountability

11. Future Scope

- Integration with digital marketplaces like **e-NAM & APMC markets**
- AI-based crop prediction with blockchain certification
- IoT device integration for real-time temperature/humidity logs
- Global export tracking for traceable Indian produce

12. Conclusion

This project offers an innovative and decentralized approach to enhance the Indian agricultural ecosystem. By using blockchain, IPFS, and QR-based traceability, the system ensures authenticity, transparency, and efficiency across the entire supply chain. It aligns with *Digital India* and *Atmanirbhar Bharat* and has the potential to revolutionize agricultural trade and governance.

13. Research and References

- **Solidity Documentation** (for blockchain smart contracts): https://soliditylang.org/

- **OpenAI ChatGPT** (for research and brainstorming support): https://chatgpt.com/

- **MDN(Mozilla developer network)**: https://developer.mozilla.org/en-US/

- **Tailwind CSS** : https://tailwindcss.com/

- **Bootstrap** : https://getbootstrap.com/

- **Font Awesome** : https://fontawesome.com/

- **CDN** : https://cdnjs.com/